

## Low Glucose Analysis - Figure Legends

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### Figure E: OFF-Cell Population Time Course (High to Low vs Normal to Low Glucose)

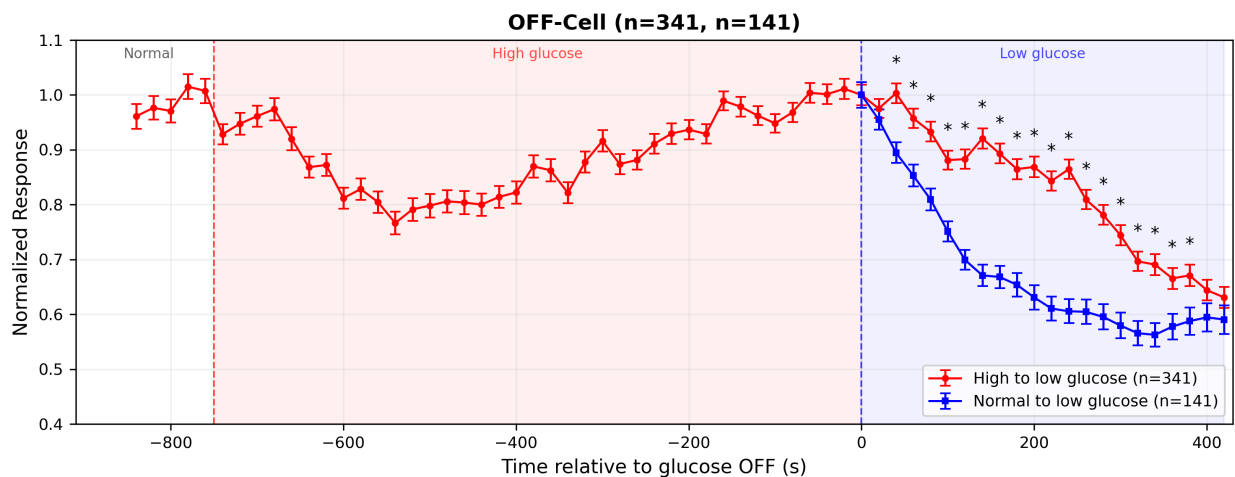


Figure Y-E2

**Figure E.** Population time course of OFF-cell light response amplitude across the full recording, with the low glucose alone control aligned at the end of high glucose ( $t=0$ ). Red trace (circles): high to low glucose – retinas exposed to normal glucose, then high glucose, then low glucose ( $n=341$  units from 5 recordings). Blue trace (squares): normal to low glucose – retinas switched directly from normal to low glucose without prior high glucose exposure ( $n=141$  units from 4 recording pairs), scaled to match the red trace at  $t=0$ . Data points show mean  $\pm$  SEM of the normalized response amplitude (max-min range), plotted trial-by-trial. Each unit's response was first normalized to its own maximum, then the population was normalized to the mean of the last 5 baseline trials before high glucose onset. Vertical dashed lines indicate solution switch times (red: high glucose onset; blue: low glucose onset / high glucose offset at  $t=0$ ). Background shading denotes solution phases: white = normal glucose, red = high glucose, blue = low glucose. Black asterisks mark time points where the two groups differ significantly (Welch's t-test,  $p < 0.05$ ). In the low glucose phase, both traces decline but the high-to-low glucose group (red) declines more slowly than the normal-to-low glucose group (blue), suggesting prior high glucose exposure partially protects against the low glucose effect on OFF-cells.

### Figure F: ON-Cell Population Time Course (High to Low vs Normal to Low Glucose)

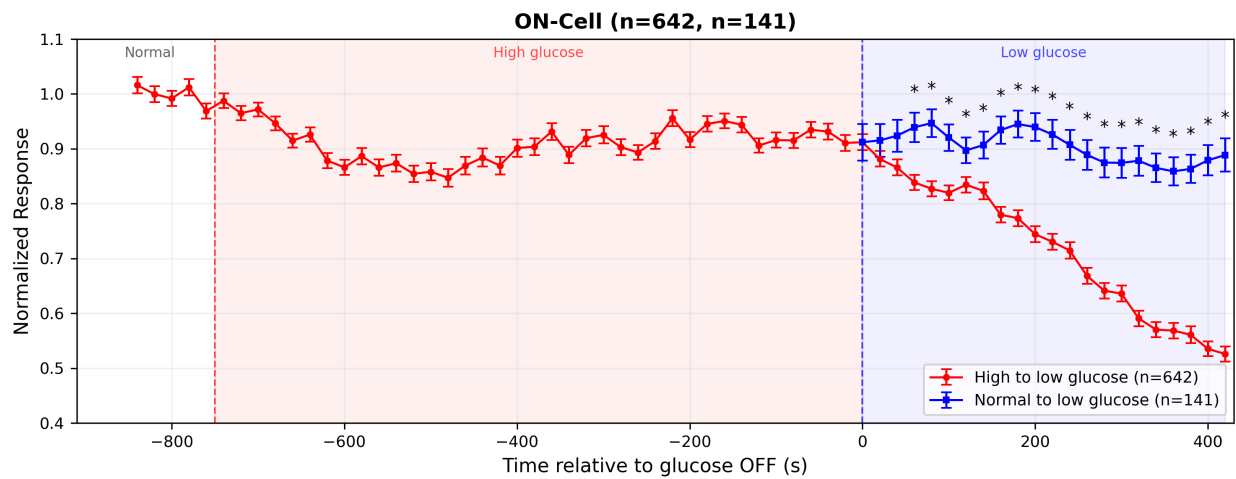


Figure Y-F2

**Figure F.** Population time course of ON-cell light response amplitude, with conventions as in Figure E. Red trace: high to low glucose (n=642 units). Blue trace: normal to low glucose (n=141 units), scaled to match the red trace at t=0. ON-cells show a similar pattern: the high-to-low glucose group (red) exhibits a more pronounced decline in the low glucose phase compared to the normal-to-low glucose group (blue), which maintains relatively stable responses. This indicates that prior high glucose exposure sensitizes ON-cells to subsequent low glucose.

## Figure S: OFF-Cell Low Glucose Phase Comparison

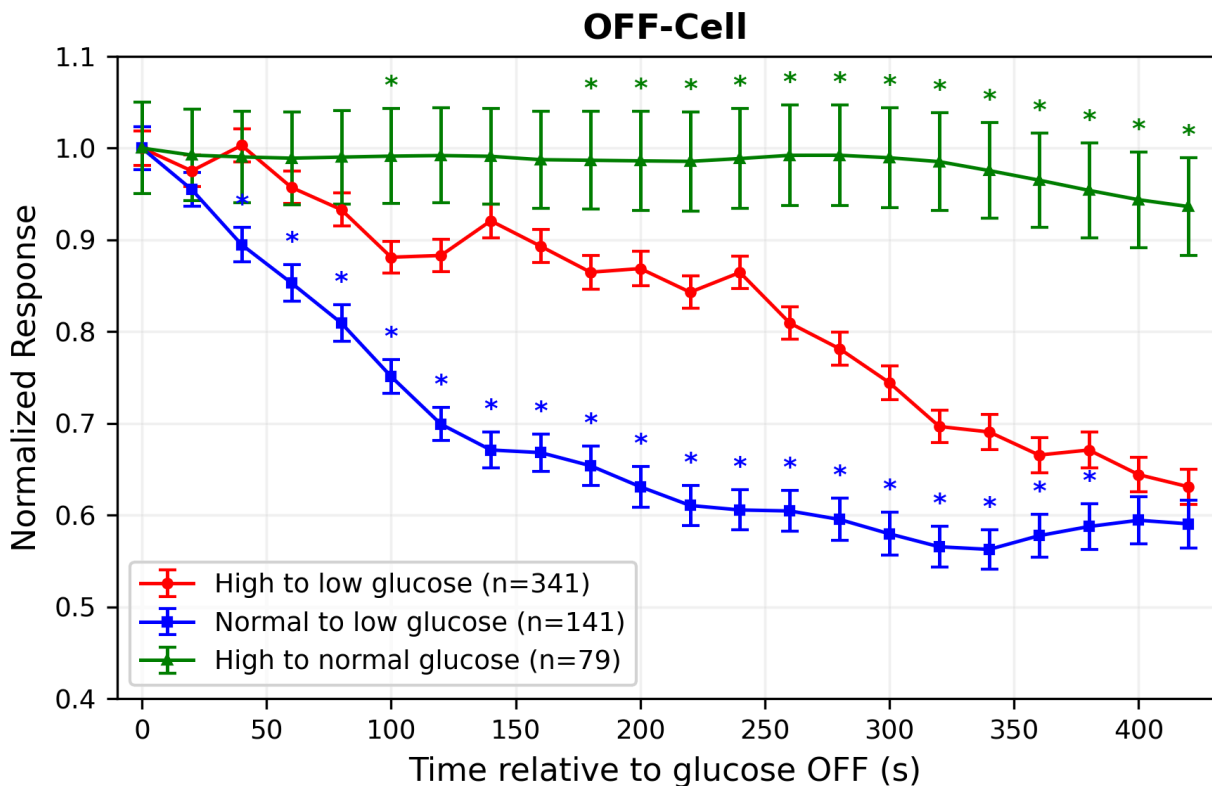


Figure Y-S

**Figure S.** Comparison of OFF-cell light response amplitude during the low glucose phase across three experimental conditions, aligned at the offset of high glucose ( $t=0$ ). Red trace (circles): high to low glucose – cells exposed to high glucose before switching to low glucose ( $n=341$  units). Blue trace (squares): normal to low glucose – cells switched directly from normal to low glucose without prior high glucose exposure ( $n=141$  units). Green trace (triangles): high to normal glucose – cells returned to normal glucose after high glucose treatment ( $n=79$  units). All traces are normalized to their respective values at  $t=0$  for direct comparison. Data points show mean  $\pm$  SEM. Colored asterisks indicate time points where the blue or green trace differs significantly from the red trace (Welch's t-test,  $p < 0.05$ ). Both low glucose conditions (red, blue) show a progressive decline in response amplitude, while the normal glucose recovery (green) remains near baseline, confirming that the response reduction is specifically caused by low glucose rather than time-dependent rundown. Notably, the decline is slower in the high-to-low glucose group (red) compared to the normal-to-low glucose group (blue), suggesting that prior high glucose exposure partially protects against the low glucose effect.

## Figure T: ON-Cell Low Glucose Phase Comparison

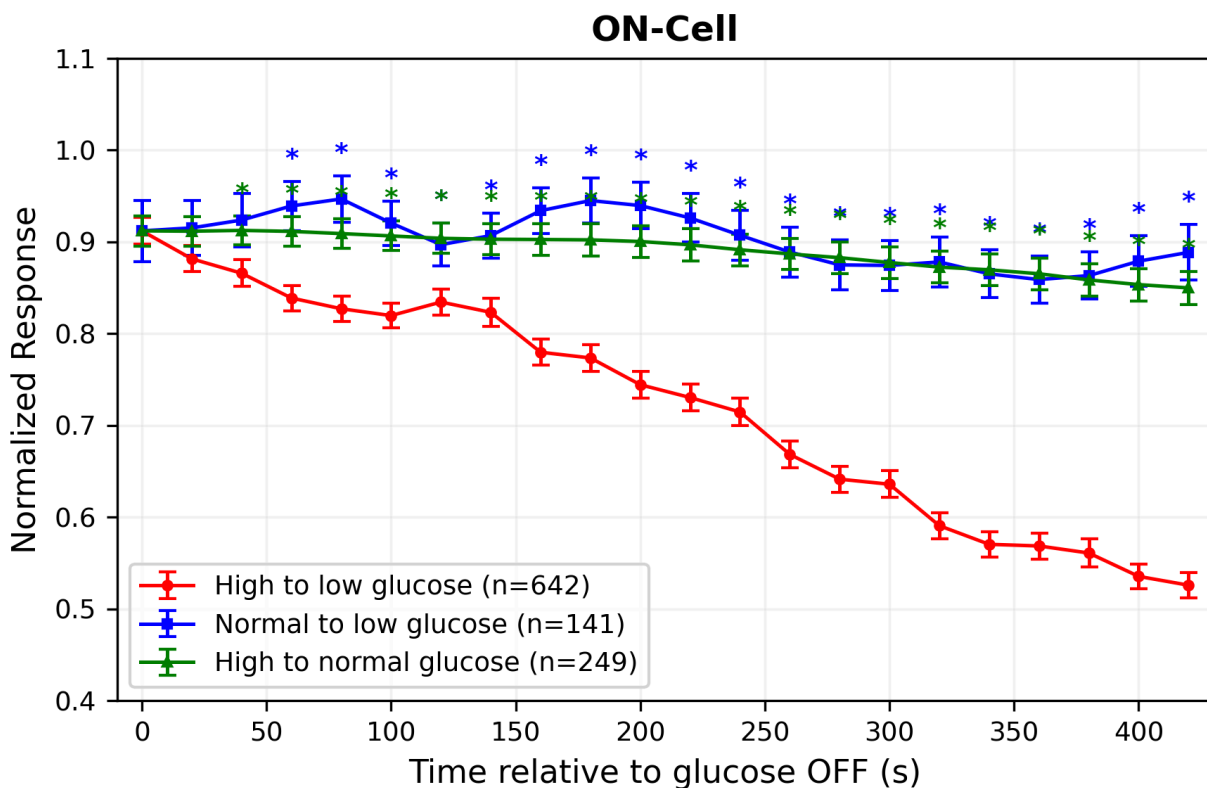


Figure Y-T

**Figure T.** Comparison of ON-cell light response amplitude during the low glucose phase across the same three conditions as Figure S. Red trace: high to low glucose ( $n=642$  units). Blue trace: normal to low glucose ( $n=141$  units). Green trace: high to normal glucose ( $n=249$  units). Conventions as in Figure S. ON-cells show a similar pattern: both low glucose conditions produce a decline in response amplitude relative to the normal glucose recovery control (green), and the high-to-low glucose group (red) declines more slowly than the normal-to-low glucose group (blue) in the early phase but ultimately reaches a lower amplitude by the end of the recording.